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Semiconductor substrate and method of dicing the same

Abstract

There is provided a method of dicing a semiconductor wafer, which includes providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing regions provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer provided on the active surface across at least a portion of the adjacent integrated circuit regions and the dicing region, forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region, propagating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7382038	12/2007	Wu	N/A	N/A
8076756	12/2010	Lane et al.	N/A	N/A
8299580	12/2011	Kumakawa et al.	N/A	N/A
8492676	12/2012	Azuma et al.	N/A	N/A
9041162	12/2014	Miccoli et al.	N/A	N/A
9406625	12/2015	Wang et al.	N/A	N/A
9640420	12/2016	Nakamura	N/A	N/A
10297710	12/2018	Kirihara	N/A	N/A
10418335	12/2018	Cho et al.	N/A	N/A
2002/0024115	12/2001	Ibnabdeljalil	257/E23.194	H01L 23/562
2003/0122220	12/2002	West	257/E23.194	H01L 23/562
2005/0272223	12/2004	Fujii	438/459	H01L 21/304
2006/0022224	12/2005	Hiroi	257/E21.582	H01L 23/522
2006/0163699	12/2005	Kumakawa	257/E23.179	H01L 23/544
2007/0066044	12/2006	Abe et al.	N/A	N/A
2007/0111477	12/2006	Maruyama et al.	N/A	N/A
2008/0067690	12/2007	Kumagai	257/774	H01L 23/585
2008/0203538	12/2007	Kumakawa et al.	N/A	N/A
2009/0014882	12/2008	Ito	257/E23.141	H01L 23/562
2009/0121313	12/2008	Hashimoto	438/667	H01L 23/585
2009/0121321	12/2008	Miccoli	257/E23.179	G03F 9/7084
2009/0121337	12/2008	Abe	257/E21.237	H01L 23/544
2009/0201043	12/2008	Kaltalioglu	324/762.05	H01L 23/585
2009/0321890	12/2008	Jeng	257/E23.193	H01L 23/562
2010/0123219	12/2009	Chen	257/E23.08	B23K 26/40
2010/0133659	12/2009	Hara	257/E23.179	H01L 23/544
2010/0283128	12/2009	Chen	257/E23.179	H01L 21/32051
2010/0314619	12/2009	Kaltalioglu	257/E21.585	H01L 22/34

2011/0241164	12/2010	Nakamura	257/774	H01L 23/585
2012/0038028	12/2011	Yaung	257/E23.18	H01L 21/78
2012/0211748	12/2011	Miccoli	257/618	H01L 21/78
2013/0075869	12/2012	Mackh	257/E23.179	H01L 21/78
2014/0264767	12/2013	Gratz	257/620	H01L 23/585
2016/0005697	12/2015	Noh	257/508	H01L 23/562
2016/0013138	12/2015	Chen	257/774	H01L 21/565
2016/0163916	12/2015	Ilievski	438/33	H01L 33/007
2016/0380605	12/2015	Matsumoto et al.	N/A	N/A
2017/0047221	12/2016	Harada	N/A	H01L 21/67092
2017/0207181	12/2016	Nakamura	N/A	H01L 21/6836
2018/0175006	12/2017	Yan	N/A	H01L 23/585
2018/0261555	12/2017	Cho	N/A	H01L 21/02675
2019/0157150	12/2018	Park	N/A	H01L 21/76811
2020/0058551	12/2019	Bae	N/A	H01L 21/6836
2021/0098392	12/2020	Wu	N/A	H01L 27/14634

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101297394	12/2007	CN	N/A
10 2018 206 303	12/2017	DE	N/A
2005-340423	12/2004	JP	N/A
2011-166183	12/2010	JP	N/A
2016-54204	12/2015	JP	N/A
2016-207765	12/2015	JP	N/A
10-2006-0020627	12/2005	KR	N/A
10-2006-0085165	12/2005	KR	N/A
10-2010-0036241	12/2009	KR	N/A
10-2017-0053112	12/2016	KR	N/A
10-2018-0104261	12/2017	KR	N/A

OTHER PUBLICATIONS

Notice of Allowance issued Jun. 29, 2025 by the Korean Patent Office for KR Patent Application No. 10-2019-0100527. cited by applicant

Communication issued on Jul. 7, 2025 by the China National Intellectual Property Administration in Chinese Patent Application No. 202010673029.1. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority from Korean Patent Application No. 10-2019-0100527, filed on Aug. 16, 2019, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

(2) The disclosure relates to a semiconductor substrate and a method of dicing the same, and more particularly, to a method of cutting a semiconductor substrate by using laser.

(3) In a chip manufacturing process, after integrated circuits are formed on an active surface of a semiconductor substrate, an inactive surface of the semiconductor substrate is polished and the polished semiconductor substrate is diced to separate the integrated circuits into respective semiconductor chips. Generally, the polished semiconductor substrate is mechanically diced using a sawing blade. When mechanical dicing is performed in this way, a diced surface of the semiconductor chips may break, causing many faults in the semiconductor chips. Hence, a method of dicing a semiconductor substrate using laser has been studied.

SUMMARY

(4) The disclosure provides a semiconductor substrate that suppresses faults generation and a method of dicing the semiconductor substrate, in a process of dicing the semiconductor substrate into semiconductor chips by using laser.

(5) Aspects of the disclosure are not limited to the foregoing, and other unmentioned aspects would be apparent to one of ordinary skill in the art from the following description.

(6) According to an aspect of the disclosure, there is provided a method of dicing a semiconductor wafer, the method comprising: providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing regions provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer provided on the active surface across at least a portion of the adjacent integrated circuit regions and the dicing region; forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region; propagating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate; and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer.

(7) According to another aspect of the disclosure, there is provided a method of dicing a semiconductor wafer, the method comprising: providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing regions provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer formed on the active surface across at least a portion of the integrated circuit regions and the dicing region; forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region; propagating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate; and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer, wherein, in a cross-sectional view, the metal shield layer includes a first metal shield layer and a second metal shield layer with a space region therebetween in a location where the crack propagates, each of the first metal shield layer and the second metal shield layer includes a major axis that is parallel to the active surface and a minor axis that is perpendicular to the active surface in the cross-sectional view, and a length of the

major axis is about 50 μm to about 100 μm, and a length of the minor axis is about 0.5 μm to about 1 μm.
(8) According to an aspect of the disclosure, there is provided a semiconductor wafer comprising: a semiconductor substrate comprising a plurality of integrated circuit regions on an active surface; a dicing regions provided between adjacent integrated circuit regions of the plurality of integrated circuit regions; and a metal shield layer provided on the active surface across a portion of the adjacent integrated circuit regions and the dicing region, wherein, in a cross-sectional view, the metal shield layer comprises a major axis that is parallel to the active surface and a minor axis that is perpendicular to the active surface, and in a planar view, the metal shield layer is arranged along a circumference of a respective integrated circuit region, among the adjacent integrated circuit regions.
(9) According to an aspect of the disclosure, there is provided a semiconductor device comprising: a semiconductor substrate comprising an integrated circuit region and a dicing region on an active surface of the semiconductor substrate, the dicing region being provided adjacent to the integrated circuit region; a metal shield layer provided on the active surface across the integrated circuit region and the dicing region; a device layer comprising one or more semiconductor devices provided on the active surface of the semiconductor substrate; one or more wires provided in a wiring layer formed on the device layer; and one or more vertical metal structures formed on the metal shield layer, wherein, in a cross-sectional view, the metal shield layer comprises a major axis that is parallel to the active surface and a minor axis that is perpendicular to the active surface, and wherein, in the cross-sectional view, the one or more vertical metal structures are perpendicular to the major axis of the metal shield layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments of the disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:
(2) FIG. **1** is a perspective view of a semiconductor substrate according to an exemplary embodiment of the disclosure;
(3) FIG. **2**A is an enlarged plane view of a region A illustrated in FIG. **1**;
(4) FIG. **2**B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **2**A in a direction X;
(5) FIG. **2**C is an enlarged plane view of a region C illustrated in FIG. **1**;
(6) FIG. **3** is a perspective view showing a state in which a protective sheet is attached to a semiconductor substrate, according to an embodiment of the disclosure;
(7) FIG. **4** is a side cross-sectional view corresponding to a line IV-IV′ illustrated in FIG. **3**, where attachment of a protective sheet to a semiconductor substrate is completed;
(8) FIG. **5**A is a perspective view showing a state where laser is irradiated into a semiconductor substrate according to an embodiment of the disclosure;
(9) FIG. **5**B is a side cross-sectional view showing a traveling direction of laser;
(10) FIG. **6** is a side cross-sectional view showing a state where irradiation of laser is completed, according to an embodiment of the disclosure;
(11) FIG. **7**A, which corresponds to FIG. **2**A, is an enlarged plane view of a region A illustrated in FIG. **6**;
(12) FIG. **7**B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **7**A in a direction X;
(13) FIG. **8** is a side cross-sectional view showing a state where a semiconductor substrate according to an embodiment of the disclosure is polished;

(14) FIG. **9**A, which corresponds to FIG. **2**A, is an enlarged plane view of a region A illustrated in FIG. **8**;
(15) FIG. **9**B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **9**A in a direction X;
(16) FIG. **10** is a side cross-sectional view showing a state where a semiconductor substrate according to an embodiment of the disclosure is diced into semiconductor chips;
(17) FIG. **11**A, which corresponds to FIG. **2**A, is an enlarged plane view of a region A illustrated in FIG. **10**;
(18) FIG. **11**B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **11**A in a direction X;
(19) FIGS. **12**A and **12**B illustrate a semiconductor substrate according to another embodiment of the disclosure, in which FIG. **12**A is an enlarged side cross-sectional view of a region A illustrated in FIG. **1** and FIG. **12**B is an enlarged plane view of a region C illustrated in FIG. **1**;
(20) FIG. **13** illustrates a semiconductor substrate according to another embodiment of the disclosure, in which FIG. **13** is an enlarged side cross-sectional view of a region A illustrated in FIG. **1**;
(21) FIG. **14** is a side cross-sectional view illustrating a semiconductor package including semiconductor chips diced from a semiconductor substrate according to an embodiment of the disclosure;
(22) FIG. **15**A is an enlarged side cross-sectional view of a portion D illustrated in FIG. **14** according to an embodiment of the disclosure;
(23) FIG. **15**B is an enlarged side cross-sectional view of a portion D illustrated in FIG. **14** according to an embodiment of the disclosure;
(24) FIG. **16** is a structural diagram showing a system of a semiconductor package including semiconductor chips diced from a semiconductor substrate according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(25) Hereinafter, embodiments of the disclosure will be described in detail with reference to the accompanying drawings.
(26) FIG. **1** is a perspective view of a semiconductor substrate according to an exemplary embodiment of the disclosure, FIG. **2**A is an enlarged plane view of a region A illustrated in FIG. **1**, FIG. **2**B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **2**A in a direction X, and FIG. **2**C is an enlarged plane view of a region C illustrated in FIG. **1**.
(27) Referring to FIG. **1**, FIGS. **2**A, **2**B and **2**C, a semiconductor substrate **100** includes integrated circuit regions **102** and a dicing region **104**.
(28) According to an exemplary embodiment, a semiconductor substrate **100** may include a wafer and may have a circular shape having a constant first thickness T**1**. The semiconductor substrate **100** may have a notch **100**N used as a reference point of wafer alignment.
(29) The semiconductor substrate **100** may include, for example, silicon. Alternatively, the semiconductor substrate **100** may include a semiconductor element like germanium or a compound semiconductor such as silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), and indium phosphide (InP). The semiconductor substrate **100** may have a silicon-on-insulator (SOI) structure. In some embodiments of the disclosure, the semiconductor substrate **100** may include a well or structure doped with impurities, which is a conductive region. The semiconductor substrate **100** may have various element isolation structures such as a shallow trench isolation (STI) structure.
(30) Herein, the semiconductor substrate **100** is assumed to have a diameter of about 12 inches, and a description will be made of a case where a silicon wafer is used. However, it will be understood by those of ordinary skill in the art that the semiconductor substrate **100** having a diameter greater or less than the above-described diameter may be used and the semiconductor substrate **100** formed

of other materials may also be used. The semiconductor substrate **100** may have a first thickness T**1** of about 0.1 mm to about 1 mm. When the first thickness T**1** of the semiconductor substrate **100** is too small, mechanical strength may be unsatisfactory; when the first thickness T**1** is too large, a time spent in subsequent grinding increases, degrading the productivity of a semiconductor chip.
(31) The semiconductor substrate **100** may include an active surface **100**F that is a front side and an inactive surface **100**B that is a back side. According to an exemplary embodiment, multiple integrated circuit regions **102** may be formed on the active surface **100**F, and the multiple integrated circuit regions **102** are to be separated into semiconductor chips (**10** of FIG. **10**).
(32) According to an exemplary embodiment, a semiconductor device SD may be formed on the active surface **100**F of the semiconductor substrate. According to an exemplary embodiment, the semiconductor substrate may be classified into a memory device and a logic device.
(33) The memory device may include a volatile memory device or a non-volatile memory device. The volatile memory device may include an existing volatile memory device and a currently developed volatile memory device, for example, dynamic random access memory (DRAM), static RAM (SRAM), thyristor RAM (TRAM), zero capacitor RAM (ZRAM), or twin transistor RAM (TTRAM). The non-volatile memory device may include an existing non-volatile memory device and a currently developed non-volatile memory device, for example, flash memory, magnetic RAM (MRAM), spin-transfer torque (STT)-MRAM, ferroelectric RAM (FRAM), phase change RAM (PRAM), resistive RAM (RRAM), nanotube RRAM, polymer RAM, nano floating gate memory, holographic memory, molecular electronics memory, or insulator resistance change memory.
(34) The logic device may be implemented with, but not limited to, a microprocessor, a graphics processor, a signal processor, a network processor, an audio codec, a video codec, an application processor, a system-on-chip, etc. The microprocessor may include, for example, a single core or a multi-core.
(35) The integrated circuit regions **102** may be arranged to be isolated from one another by the dicing region **104**. The dicing region **104** may be referred to as a scribe lane. The dicing region **104** may extend in a form of a cross in a first direction X and a second direction Y that is perpendicular to the first direction X. The dicing region **104** may have a form of a linear lane having a constant width **104**W.
(36) That is, the integrated circuit regions **102** may be arranged to be separated from one another by being surrounded by the dicing region **104** in all directions. As will be described below, as the semiconductor substrate **100** and various types of material layers formed on the semiconductor substrate **100** are diced through a dicing process performed along the dicing region **104**, the integrated circuit regions **102** may be separated into a plurality of semiconductor chips (**10** of FIG. **10**) from one another.
(37) A semiconductor device layer **110** may be formed on the active surface **100**F of the semiconductor substrate **100**. The semiconductor device layer **110** may correspond to a plurality of semiconductor devices (SDs) in the integrated circuit regions **102** and may correspond to a region including a plurality of semiconductor dummy devices in the dicing region **104**.
(38) Multi-layer wires **120** may be formed from a top surface of the semiconductor device layer **110** to a bottom surface of an upper material film **130**. The multi-layer wires **120** may include an inter-layer insulating film **124** and a metal wire **126** that are arranged alternately. The multi-layer wires **120** may include a plurality of metal vertical structures **122** arranged in a third direction Z that is perpendicular to the active surface **100**F of the semiconductor substrate **100**.
(39) The metal wire **126** may include a conductive material including at least one of aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt), and gold (Au), and the plurality of metal vertical structures **122** may include a material that is substantially the same as the metal wire **126**.
(40) The inter-layer insulating film **124** may include a low dielectric material. The low dielectric material, which has a lower dielectric constant than that of a silicon oxide, may be useful for high

integration and high speed of the semiconductor device SD when the low dielectric material is used as the inter-layer insulating film **124** in the semiconductor device SD.
(41) In some exemplary embodiments of the disclosure, the inter-layer insulating film **124** may be formed to have a structure in which a first inter-layer insulating film, a second inter-layer insulating film, and a third inter-layer insulating film are sequentially stacked on the metal wire **126**. However, the number of films of the inter-layer insulating film **124** is not limited to the above-described example. The inter-layer insulating film **124** may be formed to fill peripheries of the plurality of metal vertical structures **122** and the metal wire **126** that are formed of a conductive material.
(42) According to an exemplary embodiment, opposite side walls of each of the plurality of metal vertical structures **122** may be formed flat. Each of the plurality of metal vertical structures **122** may be formed of single metal, may have flat side walls, and may have a bar shape with a major axis in the third direction Z and a minor axis in the first direction X. Thus, the plurality of metal vertical structures **122** do not include a bonding interface and a crack producing portion between heterogeneous materials, thereby properly blocking the propagation of a crack described below (CR of FIG. **9**B) in the first direction X and the second direction Y and thus effectively inducing the propagation of the crack (CR of FIG. **9**B) in the third direction Z.
(43) In the dicing region **104**, the multi-layer wires **120** may be formed as a multi-layer dummy wire structure corresponding to metal wires formed in the integrated circuit regions **102**.
(44) Although the multi-layer wires **120** are shown as three layers, they are not limited thereto. For example, the multi-layer wires **120** may be formed as two layers or four or more layers.
(45) The upper material film **130** may be formed on the multi-layer wires **120**. That is, a level of the lowermost surface of the upper material film **130** may be the same as or higher than a level of the uppermost surface of the multi-layer wires **120**.
(46) The upper material film **130** may be formed in a form where a first material film **132**, a second material film **134**, and a third material film **136** are sequentially stacked. Each of the first material film **132**, second material film **134**, and third material film **136** may include an insulating film and may be formed of different materials. In some exemplary embodiments of the disclosure, the first material film **132** may include a silicon oxide, the second material film **134** may include a silicon carbonitride (SiCN), and the third material film **136** may include a silicon nitride. Although the upper material film **130** is shown as three films, it is not limited thereto. For example, unlike shown, the upper material film **130** may be formed as two films or four or more films.
(47) In other exemplary embodiments of the disclosure, the upper material film **130** may be formed to have a structure where a silicon oxide, such as phosphor silicate glass (PSG), boro-phosphor silicate glass (BPSG), undoped silicate glass (USG), tetra ethyl ortho silicate (TEOS), plasma enhanced-TEOS (PE-TEOS), or high density plasma-chemical vapor deposition (HDP-CVD) oxide, and a silicon nitride are stacked alternately.
(48) According to another exemplary embodiment, the upper material film **130** of the dicing region **104** may include a test pattern for testing electrical characteristics of the semiconductor device SD existing in the integrated circuit regions **102**, a redistribution layer for electrical connection between the test patterns, or an alignment key for aligning a mask.
(49) According to an exemplary embodiment of the disclosure, a protective film **140** may be formed to expose the upper material film **130** located in the dicing region **104**W and to cover the upper material film **130** located in the integrated circuit regions **102**. A side wall of the protective film **140** may be an inclined plane. The protective film **140** may include a material film formed of, for example, an organic compound. In some exemplary embodiments of the disclosure, the protective film **140** may include a material film formed of an organic high-polymer material. In other exemplary embodiments of the disclosure, the protective film **140** may include photosensitive polyimide (PSPI) resin. A width **104**W of the dicing region **104** exposed from the protective film **140** may be about 5 μm to about 100 μm. However, a numerical value of the width **104**W is not

limited to the above example.
(50) The metal shield layer MS may be formed on the active surface **100**F of the semiconductor substrate **100** across the integrated circuit regions **102** and the dicing region **104**. The metal shield layer MS may prevent a spot generated due to leakage or scattering of laser in a dicing process from spreading to the integrated circuit regions **102**.
(51) A bottom surface of the metal shield layer MS may be arranged to directly contact the active surface **100**F. In other words, the metal shield layer MS may be formed in a middle-end-of-line (MEOL) process. Thus, in the integrated circuit regions **102** of the semiconductor substrate **100**, the semiconductor device SD may not be arranged in a location where the metal shield layer MS is formed.
(52) In a side cross-section view, the metal shield layer MS may include a first metal shield layer MS**1** and a second metal shield layer MS**2** with a space region SS of a first interval therebetween in the dicing region **104**. In terms of locations, the first metal shield layer MS**1** may be referred to as a left metal shield layer and the second metal shield layer MS**2** may be referred to as a right metal shield layer. The first interval of the space region SS may be narrower than the width **104**W of the dicing region **104**.
(53) The first metal shield layer MS**1** may have a major axis MS**1**R in the first direction X that is parallel to the active surface **100**F and a minor axis MS in the third direction Z that is perpendicular to the active surface **100**F. The second metal shield layer MS**2** may have a major axis MS**2**R in the first direction X that is parallel to the active surface **100**F and a minor axis MS**2**S in the third direction Z that is perpendicular to the active surface **100**F.
(54) Lengths of the major axes MS**1**R and MS**2**R may be about 50 μm to about 100 μm, and lengths of the minor axes MS**1**S and MS**2**S may be about 0.5 μm to about 1 μm. Describing the above examples as ratios, a ratio of the lengths of the major axes MS**1**R and MS**2**R to the lengths of the minor axes MS and MS**2**S may be about 50:1 to about 200:1.
(55) That is, each of the first metal shield layer MS**1** and the second metal shield layer MS**2** may be formed as a thin flat plate-type structure to cover the semiconductor substrate **100**. The first metal shield layer MS**1** and the second metal shield layer MS**2** may substantially have the same shape as each other, but the disclosure is not limited thereto.
(56) In a side cross-section view, a propagation direction of a crack (CR of FIG. **9**B) and a major-axial direction of the metal shield layer MS may be perpendicular to each other. That is, the propagation direction of the crack (CR of FIG. **9**B) may be the third direction Z, and the major-axial direction of the metal shield layer MS may be the first direction X.
(57) In a plane view, the metal shield layer MS may include single metal in a hollow rectangular or square shape in each of adjacent integrated circuit regions **102** with the dicing region **104** therebetween. To allow the crack (CR of FIG. **9**B) to pass through the space region SS, the metal shield layers MS may be arranged to be spaced apart from each other by a distance. In other words, the metal shield layer MS may be arranged to surround a semiconductor chip (**10** of FIG. **10**).
(58) A material of the metal shield layer MS may be metal having a melting point of about 600° C. or higher. To prevent the metal shield layer MS from being melted, the metal shield layer MS may include metal having a melting point higher than a temperature of a portion of the semiconductor substrate **100** heated by laser. In some exemplary embodiments of the disclosure, the metal shield layer MS may include aluminum (a melting point of about 660° C.).
(59) Recently, as large volume and high integration of a semiconductor device are required, an area occupied by a dicing region in a semiconductor substrate is reduced. Generally, the semiconductor substrate is mechanically diced using a sawing blade. As such, when mechanical dicing is performed, the risk of damage to integrated circuit regions may increase due to stress applied to the semiconductor substrate during a dicing process.
(60) Hence, a process of dicing a semiconductor substrate by using laser has been performed. However, due to a difference between a density of a modified layer of the semiconductor substrate

and a density of a periphery of the modified layer, a part of laser that has to be focused onto an inside of the semiconductor substrate leaks or is scattered, and penetrates a semiconductor device of an integrated circuit region, causing a fault.
(61) Thus, the semiconductor substrate **100** according to the disclosure may prevent the leaking or scattering laser from entering regions where the semiconductor device SD are located by forming the metal shield layer MS across the integrated circuit regions **102** and the dicing region **104** of the semiconductor substrate **100**. In this way, it is possible to prevent a fault such as a function failure occurring in the semiconductor device SD due to leaking or scattering of laser.
(62) As such, a semiconductor chip diced from the semiconductor substrate **100** according to the disclosure has less faults in a dicing process, thereby improving electrical characteristics and production efficiency of the semiconductor chip.
(63) Hereinbelow, a method of dicing the semiconductor substrate **100** including the metal shield layer MS will be described in detail.
(64) FIG. **3** is a perspective view showing a state in which a protective sheet is attached to a semiconductor substrate, according to an exemplary embodiment of the disclosure, and FIG. **4** is a side cross-sectional view corresponding to a line IV-IV′ illustrated in FIG. **3**, where attachment of a protective sheet to a semiconductor substrate is completed.
(65) Referring to FIGS. **3** and **4**, a protective sheet **200** may be attached onto the active surface **100**F of the semiconductor substrate **100**.
(66) The protective sheet **200** may protect the integrated circuit regions **102** during a dicing process of the semiconductor substrate **100**.
(67) The protective sheet **200** may include a polymer sheet based on polyvinylchloride (PVC) and may be attached onto the active surface **100**F using an acrylic resin-based adhesive. The acrylic resin-based adhesive may have a thickness of about 2 μm to about 10 μm, and the protective sheet **200** may have a thickness of about 60 μm to about 200 μm. The protective sheet **200** may have a circular shape having substantially the same diameter as a diameter of the semiconductor substrate **100**.
(68) FIG. **5**A is a perspective view showing a state where laser is irradiated into a semiconductor substrate according to an exemplary embodiment of the disclosure, and FIG. **5**B is a side cross-sectional view showing a traveling direction of laser.
(69) Referring to FIGS. **5**A and **5**B, after the protective sheet **200** is attached onto the active surface **100**F of the semiconductor substrate **100**, laser RA of a wavelength having penetrability with respect to the semiconductor substrate **100** may be controlled to have a converging point **150**P inside the semiconductor substrate **100** and thus may be irradiated along the dicing region **104**.
(70) When the laser RA is irradiated to an inside of the semiconductor substrate **100**, a modified layer (**150** of FIG. **6**) may be formed along the dicing region **104** inside the semiconductor substrate **100**. The modified layer (**150** of FIG. **6**) may be formed using a laser irradiation apparatus **300**.
(71) The laser irradiation apparatus **300** may include a chuck table **310** supporting the semiconductor substrate **100**, a laser irradiation device **320** irradiating the laser RA to the semiconductor substrate **100** arranged on the chuck table **310**, and an imaging device **330** imaging the semiconductor substrate **100** arranged on the chuck table **310**. The chuck table **310** may support the semiconductor substrate **100** through suction using vacuum pressure and may move in the first direction X and the second direction Y.
(72) The laser irradiation device **320** may be configured to irradiate pulse laser from a condenser **324** mounted on a front end of a housing **322** in a cylindrical shape arranged substantially horizontally. While the condenser **324** irradiates pulse laser of a wavelength having penetrability with respect to the semiconductor substrate **100**, the chuck table **310** and the condenser **324** may perform relative movement at a proper speed.
(73) The imaging device **330** mounted on another front end of the housing **322** constituting the

laser irradiation device **320** may be a general charge-coupled device (CCD) imaging device that performs imaging by using visible rays. In another exemplary embodiment of the disclosure, the imaging device **330** may include an infrared irradiation device that irradiates infrared rays to the semiconductor substrate **100**, an optical system that captures the infrared rays irradiated by the infrared irradiation rays, and an infrared CCD imaging device that outputs an electrical signal corresponding to the infrared rays captured by the optical system.
(74) The laser irradiation device **320** may irradiate the laser RA after being aligned in a laser irradiation location. The converging point **150**P of the laser RA may be controlled to be located closer to the active surface **100**F than to the inactive surface **100**B of the semiconductor substrate **100**. That is, the modified layer (**150** of FIG. **6**) may be located closer to the active surface **100**F.
(75) The laser RA emitted from the laser irradiation device **320** may be intensively irradiated such that a part of the semiconductor substrate **100** is heated to a temperature of about 600° C. That is, the part of the semiconductor substrate **100** located in the converging point **150**P of the laser RA may be melted by the laser RA.
(76) As such, when the part of the semiconductor substrate **100** is melted, a crystalline state of the part changes, causing unintended leakage or scattering RB of the laser RA. The leakage or scattering RB of the laser RA may cause a spot in a part other than the converging point **150**P of the semiconductor substrate **100**, and the spot may cause a fault of the semiconductor device.
(77) FIG. **6** is a side cross-sectional view showing a state where irradiation of laser is completed, according to an exemplary embodiment of the disclosure, FIG. **7**A, which corresponds to FIG. **2**A, is an enlarged plane view of a region A illustrated in FIG. **6**, and FIG. **7**B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **7**A in a direction X.
(78) Referring to FIGS. **6**, **7**A and **7**B, the modified layer **150** may be located away from the inactive surface **100**B of the semiconductor substrate **100** by a first distance D**1**, and the modified layer **150** may be located closer to the active surface **100**F.
(79) Laser may be easily irradiated to a target location through light amplification by stimulated emission radiation. By using properties of the laser, the modified layer **150** may be formed in a target location inside the semiconductor substrate **100**. The modified layer **150** may include a crack site where the crack (CR of FIG. **9**B) is initiated by an external physical impact.
(80) A first spot **151** and a second spot **153** may be formed due to unintended leakage or scattering of laser. The first spot **151** may indicate a spot formed inside the semiconductor substrate **100**, and the second spot **153** may indicate a spot formed inside the metal shield layer MS. The second spot **153** is formed in a location that may substantially affect the semiconductor device SD, but the semiconductor substrate **100** according to the disclosure may solve such a problem by including the metal shield layer MS.
(81) The modified layer **150** may be located under the semiconductor device layer **110**. The modified layer **150** has a constant width in the first direction X, and a virtual line connecting the modified layers **150** in the second direction Y may have a linear lane form.
(82) FIG. **8** is a side cross-sectional view showing a state where a semiconductor substrate according to an exemplary embodiment of the disclosure is polished, FIG. **9**A, which corresponds to FIG. **2**A, is an enlarged plane view of a region A illustrated in FIG. **8**, and FIG. **9**B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **9**A in a direction X.
(83) Referring to FIGS. **8**, **9**A and **9**B, the crack CR may be initiated in the dicing region **104** by polishing the inactive surface **100**B of the semiconductor substrate **100**.
(84) By polishing the inactive surface **100**B of the semiconductor substrate **100** using a polishing device **400**, a thickness of the semiconductor substrate **100** may be reduced and the crack CR may propagate from the modified layer **150**.
(85) The polishing device **400** may include a chuck table **410** that supports the semiconductor substrate **100** and a grinder **420** that polishes the semiconductor substrate **100** arranged on the

chuck table **410**. The grinder **420** may move while rotating, and a polishing pad may be attached under the grinder **420**.
(86) The semiconductor substrate **100**, which is polished, may have a second thickness T**2** that is substantially less than the initial first thickness (T**1** of FIG. **2**B). The second thickness T**2** may be about 20 μm to about 50 μm.
(87) By polishing the semiconductor substrate **100** using the polishing device **400**, the semiconductor substrate **100**, which is polished, having a final thickness may be formed. At the same time, in the dicing region **104**, the crack CR may pass through the active surface **100**F of the polished semiconductor substrate **100** from the modified layer **150** and propagate in the third direction Z away from the active surface **100**F.
(88) According to an exemplary embodiment of the disclosure, laser may be irradiated to the inside of the semiconductor substrate **100** to form the modified layer **150** along the dicing region **104** of the semiconductor substrate **100**, and then the inactive surface **100**B of the semiconductor substrate **100** may be polished. A polishing process may be a grinding process in a state where physical pressure is applied to the semiconductor substrate **100**.
(89) When the polishing process is performed in the state where the physical pressure is applied to the semiconductor substrate **100**, the polished semiconductor substrate **100** may be brittle and fractured. The brittle fracture indicates that when a force greater than or equal to an elastic limit is applied to an object, the object is fractured instead of permanently deformed. Thus, during polishing of the inactive surface **100**B of the semiconductor substrate **100**, the semiconductor substrate **100** that is gradually thinner may be brittle fractured by the crack CR propagating from the modified layer **150**. As the crack CR propagating from the modified layer **150** is initiated along the dicing region **104** isolating the integrated circuit regions **102**, the integrated circuit regions **102** may be separated into semiconductor chips (**10** of FIG. **10**) by the brittle fracture of the semiconductor substrate **100**. The semiconductor chip (**10** of FIG. **10**) may be fixed in an original location without leaving the original location, by the protective sheet **200**.
(90) In other exemplary embodiments of the disclosure, by continuing polishing the inactive surface **100**B of the semiconductor substrate **100**, the modified layer **150** may be completely removed. A diced surface of the semiconductor chip (**10** of FIG. **10**) separated with complete removal of the modified layer **150** may be smoother than a diced surface that is mechanically diced using a sawing blade. Moreover, by completely removing the modified layer **150** in the polishing process, the crack site in the modified layer **150** may be entirely removed, without causing another crack CR.
(91) The crack CR propagating from the modified layer **150** may propagate in the third direction Z. Even when the crack CR partially propagates in the first direction X and/or the second direction Y, the crack CR may not propagate to the integrated circuit regions **102** by being blocked by the plurality of metal vertical structures **122**.
(92) Moreover, by using laser, a cut width of the semiconductor substrate **100** may be reduced. Thus, the width of the dicing region **104** may be reduced as compared to mechanical dicing using a sawing blade, forming more integrated circuit regions **102** on the semiconductor substrate **100**.
(93) FIG. **10** is a side cross-sectional view showing a state where a semiconductor substrate according to an exemplary embodiment of the disclosure is diced into semiconductor chips, FIG. **11**A, which corresponds to FIG. **2**A, is an enlarged plane view of a region A illustrated in FIG. **10**, and FIG. **11**B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **11**A in a direction X.
(94) Referring to FIGS. **10**, **11**A and **11**B together, the semiconductor substrate **100** may be separated into the semiconductor chips **10** by a dicing process.
(95) More specifically, in the semiconductor substrate **100**, the integrated circuit regions **102** may be separated into the semiconductor chips **10** by the crack (CR of FIG. **9**B) in the dicing region **104**. The separated semiconductor chips (**10** of FIG. **10**) may be fixed in an original location

without leaving the original location, by the protective sheet **200**.
(96) A method of dicing the semiconductor substrate **100** according to the disclosure may form the metal shield layer MS across the integrated circuit regions **102** and the dicing region **104** of the semiconductor substrate **100**, thereby preventing leaking or scattered laser from affecting the semiconductor device and reducing the width of the dicing region **104**.
(97) In other words, in the method of dicing the semiconductor substrate **100** using laser, an influence of leakage or scattering of the laser may be removed by the metal shield layer MS, thereby preventing a fault such as a function failure of the semiconductor device.
(98) Eventually, a fault of the semiconductor chip **10** diced by the method of dicing the semiconductor substrate **100** may be reduced, and electrical characteristics and productivity of the semiconductor chip **10** may be improved.
(99) FIGS. **12**A and **12**B illustrate a semiconductor substrate according to another exemplary embodiment of the disclosure, in which FIG. **12**A is an enlarged side cross-sectional view of a region A illustrated in FIG. **1** and FIG. **12**B is an enlarged plane view of a region C illustrated in FIG. **1**.
(100) Hereinbelow, many components of a semiconductor substrate **100-1** and materials of the components are substantially the same as or similar to a description made with reference to FIGS. **1**, **2**A, **2**B and **2**C. Thus, for convenience, a description will be described based on differences than the semiconductor substrate (**100** of FIG. **1**) described above.
(101) Referring to FIGS. **12**A and **12**B, the semiconductor substrate **100-1** may include the metal shield layer MS continuing across the adjacent integrated circuit regions **102** with the dicing region **104** therebetween.
(102) The metal shield layer MS may be formed on the active surface **100**F of the semiconductor substrate **100** across the adjacent integrated circuit regions **102** and the dicing region **104**. The metal shield layer MS may prevent a spot generated due to leakage or scattering of laser in a dicing process from spreading to the integrated circuit regions **102**.
(103) A bottom surface of the metal shield layer MS may be arranged to directly contact the active surface **100**F. In other words, the metal shield layer MS may be formed in an MEOL process. Thus, in the integrated circuit regions **102** of the semiconductor substrate **100**, the semiconductor device SD may not be arranged in a location where the metal shield layer MS is formed.
(104) In a side cross-section view, the metal shield layer MS may have a major axis MSR in the first direction X that is parallel to the active surface **100**F and a minor axis MSS in the third direction Z that is perpendicular to the active surface **100**F.
(105) A length of the major axis MSR may be about 100 μm to about 200 μm, and a length of the minor axis MSS may be about 0.5 μm to about 1 μm. Describing the above example as a ratio, a ratio of the length of the major axis MSR to the length of the minor axis MSS may be about 100:1 to about 400:1.
(106) That is, the metal shield layer MS may be formed as a thin flat plate-type structure to cover the semiconductor substrate **100**.
(107) In a side cross-section view, a propagation direction of a crack (CR of FIG. **9**B) and a major-axial direction of the metal shield layer MS may be perpendicular to each other. That is, the propagation direction of the crack (CR of FIG. **9**B) may be the third direction Z, and the major-axial direction of the metal shield layer MS may be the first direction X. In a dicing process, the crack (CR of FIG. **9**B) may penetrate a central portion of the metal shield layer MS located in the dicing region **104**.
(108) In a plane view, the metal shield layer MS may include integrated metal in a hollow lattice shape in each of adjacent integrated circuit regions **102**, covering the active surface **100**F corresponding to the dicing region **104**. The metal shield layer MS may be arranged to allow the crack (CR of FIG. **9**B) to pass through the metal shield layer MS. In other words, the metal shield layer MS may be arranged to surround a semiconductor chip (**10** of FIG. **10**).

(109) A material of the metal shield layer MS may be metal having a melting point of about 600° C. or higher. To prevent the metal shield layer MS from being melted, the metal shield layer MS may include metal having a melting point higher than a temperature of a portion of the semiconductor substrate **100** heated by laser. In some exemplary embodiments of the disclosure, the metal shield layer MS may include aluminum (a melting point of about 660° C.).
(110) FIG. **13** illustrates a semiconductor substrate according to another exemplary embodiment of the disclosure, in which FIG. **13** is an enlarged side cross-sectional view of a region A illustrated in FIG. **1**;
(111) Hereinbelow, many components of a semiconductor substrate **100-2** and materials of the components are substantially the same as or similar to a description made with reference to FIGS. **1** through **2**C. Thus, for convenience, a description will be described based on differences than the semiconductor substrate (**100** of FIG. **1**) described above.
(112) Referring to FIG. **13**, a semiconductor substrate **100-2** may include the integrated circuit regions **102** and the dicing region **104**, without including a plurality of metal vertical structures.
(113) The multi-layer wires **120** may be formed from a top surface of the semiconductor device layer **110** to a bottom surface of an upper material film **130**. The multi-layer wires **120** may include an inter-layer insulating film **124** and a metal wire **126** that are arranged alternately.
(114) The metal wire **126** may include a conductive material including at least one of aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt), and gold (Au).
(115) The inter-layer insulating film **124** may include a low dielectric material. The low dielectric material, which has a lower dielectric constant than that of a silicon oxide, may be useful for high integration and high speed of the semiconductor device SD when the low dielectric material is used as the inter-layer insulating film **124** in the semiconductor device SD.
(116) In some exemplary embodiments of the disclosure, the inter-layer insulating film **124** may be formed to have a structure in which a first inter-layer insulating film, a second inter-layer insulating film, and a third inter-layer insulating film are sequentially stacked with the metal wire **126** thereamong. However, the number of films of the inter-layer insulating film **124** is not limited to the above-described example. The inter-layer insulating film **124** may be formed to fill the peripheries of the plurality of metal vertical structures **122** and the metal wire **126**.
(117) The metal shield layer MS may be formed on the active surface **100**F of the semiconductor substrate **100** across the integrated circuit regions **102** and the dicing region **104**. The metal shield layer MS may prevent a spot generated due to leakage or scattering of laser in a dicing process from spreading to the integrated circuit regions **102**.
(118) In a side cross-section view, the metal shield layer MS may include a first metal shield layer MS**1** and a second metal shield layer MS**2** with a space region SS of a first interval therebetween in the dicing region **104**. In terms of locations, the first metal shield layer MS**1** may be referred to as a left metal shield layer and the second metal shield layer MS**2** may be referred to as a right metal shield layer. The first interval of the space region SS may be narrower than the width **104**W of the dicing region **104**.
(119) FIG. **14** is a side cross-sectional view illustrating a semiconductor package including semiconductor chips diced from a semiconductor substrate according to an exemplary embodiment of the disclosure, FIG. **15**A is an enlarged side cross-sectional view of a portion D illustrated in FIG. **14** according to an exemplary embodiment of the disclosure, and FIG. **15**B is an enlarged side cross-sectional view of a portion D illustrated in FIG. **14** according to an exemplary embodiment of the disclosure.
(120) Referring to FIG. **14**, a semiconductor package **1000** may include a package substrate **1010**, the plurality of semiconductor chips **10**, a connection member **1020**, and a molding member **1030**.
(121) The package substrate **1010** as a support substrate may include a body **1011**. According to exemplary embodiment, the package substrate **1010** may further include a lower protective layer, and an upper protective layer. The package substrate **1010** may be formed based on a printed circuit

board (PCB), a wafer substrate, a ceramic substrate, a glass substrate, an interposer substrate, etc. In an exemplary embodiment of the disclosure, the package substrate **1010** may be a PCB. The package substrate **1010** is not limited to a PCB.

(122) According to an embodiment, a wire **1016** may be formed on the package substrate **1010**, and the wire **1016** may be electrically connected to the semiconductor chip **10** through the connection member **1020** connected to the upper electrode pad **1012** on a top surface of the package substrate **1010**. An external connection terminal **1040** may be arranged on the lower electrode pad **1014** on a bottom surface of the package substrate **1010**. The package substrate **1010** may be mounted in electrical connection with a module substrate or a system board of an electronic product through the external connection terminal **1040**.

(123) The multi-layer or single-layer wire, for example, the wire **1016**, may be formed in the body **1011**, and the semiconductor chip **10** may be electrically connected with the external connection terminal **1040** through the wire **1016**. The lower protective layer and the upper protective layer may function to protect the body **1011**, and may be formed as, for example, solder resist.

(124) When the package substrate **1010** is a PCB, the body **1011** may be generally implemented by compressing a high-polymer material such as thermoset resin, epoxy-based resin such as flame retardant **4** (FR-4), bismaleimide triazine (BT), an Ajinomoto build up film (ABF), etc., phenol resin, etc., to form a thin film, applying copper foil to opposite sides, and forming the wire **1016** that is a delivery path of an electrical signal through patterning. Except for parts connected with a terminal, e.g., the upper electrode pad **1012** and the lower electrode pad **1014**, solder resist may be coated onto a bottom surface and a top surface of the body **1011**, thus implementing the lower protective layer and the upper protective layer.

(125) Meanwhile, the PCB may be divided into a single layer PCB in which the wire **1016** is formed only on a single side, and a double layer PCB in which the wire **1016** is formed on opposite sides. The number of layers of the copper foil may be three or more by using an insulator called prepreg, and three or more wires **1016** may be formed based on the number of formed copper foils, thus implementing a PCB having a multi-layer structure. However, the package substrate **1010** is not limited to the above-described structure or material of the PCB.

(126) The plurality of semiconductor chips **10** may be electrically connected to the package substrate **1010** by the connection member **1020**. The connection member **1020** may electrically connect the package substrate **1010** and the plurality of semiconductor chips **10** by electrically connecting the upper electrode pad **1012** of the package substrate **1010** with the connection pad **12** of the semiconductor chip **10**. In some exemplary embodiments of the disclosure, the connection member **1020** may be a bonding wire.

(127) The connection member **1020** may be used to electrically connect the semiconductor chip with the package substrate **1010**. Through the connection member **1020**, at least one of a control signal, a power signal, and a ground signal for an operation of the semiconductor chip **10** may be provided from outside, a data signal to be stored in the semiconductor chip **10** may be provided from outside, or data stored in the semiconductor chip **10** may be provided to outside.

(128) The molding member **1030** may protect the plurality of semiconductor chips **10** from an external environment by surrounding the plurality of semiconductor chips **10**. The molding member **1030** may be used to inject a proper amount of molding resin onto the package substrate **1010** through an injection process and form an appearance of the semiconductor package **1000** through a hardening process. Depending on a need, in a pressing process, the appearance of the semiconductor package **1000** may be formed by applying pressure to the molding resin. Herein, a processing condition such as a delay time between injection of the molding resin and the pressing, the amount of injection molding resin, and the pressing temperature/pressure, etc., may be set based on physical properties such as viscosity of molding resin.

(129) A side surface and a top surface of the molding member **1030** may have a right-angled shape. In a process of manufacturing the semiconductor package **1000** by dicing the package substrate

1010 along a dicing line, the side surface and the top surface of the molding member **1030** may generally have a right-angled shape. Although not shown, a marking pattern including information of the semiconductor chip **10**, e.g., a barcode, a number, a character, a symbol, etc., may be formed in a part of the side surface of the semiconductor package **1000**.
(130) In some exemplary embodiments of the disclosure, the molding resin may include epoxy-group molding resin, polyimide-group molding resin, etc. The molding member **1030** may include, for example, an epoxy molding compound (EMC).
(131) The connection pad **12** may be arranged on a semiconductor device layer, and may be electrically connected with a wire layer inside the semiconductor device layer. The wire layer may be electrically connected with the connection member **1020** through the connection pad **12**. The connection pad **12** may include at least one of aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt), and gold (Au).
(132) On the semiconductor device layer may be formed a passivation layer for protecting the semiconductor device layer, the wire layer, and other structures from an external shock or moisture. The passivation layer may expose at least a part of the top surface of the connection pad **12**.
(133) The plurality of semiconductor chips **10** constituting the semiconductor package **1000** may have a stacked structure. When eight semiconductor chips **10** are stacked as shown in the figures, four semiconductor chips **10** may form a group and the semiconductor package **1000** may include two groups.
(134) When the four semiconductor chips **10** of a first group are stacked from the bottom layer to the top layer one by one, the semiconductor chips **10** may be stepwisely moved for arrangement in the first direction X to expose the connection pad **12** arranged in each semiconductor chip **10**.
(135) The four semiconductor chips **10** of a second group may be stacked on the first group. The four semiconductor chips **10** of the second group may be stepwisely moved for arrangement in the first direction X opposite to a direction in which the first group is moved.
(136) Each semiconductor chip **10** constituting the semiconductor package **1000** may include a semiconductor chip diced from the semiconductor substrate according to the disclosure. In some exemplary embodiments of the disclosure, a semiconductor chip diced from the semiconductor substrate **100** described above may be included as shown in FIG. **15**A. In other exemplary embodiments of the disclosure, a semiconductor chip diced from the semiconductor substrate **100-1** described above may be included as shown in FIG. **15**B.
(137) FIG. **16** is a structural diagram showing a system of a semiconductor package including semiconductor chips diced from a semiconductor substrate according to an exemplary embodiment of the disclosure.
(138) Referring to FIG. **16**, a system **1100** may include a controller **1110**, an input/output device **1120**, a memory **1130**, an interface **1140**, and a bus **1150**.
(139) The system **1100** may be a mobile system or a system that transmits or receives information. In some exemplary embodiments of the disclosure, the mobile system may be a portable computer, a web tablet, a mobile phone, a digital music player, or a memory card.
(140) The controller **1110** is intended to control an execution program in the system **1100**, and may include a microprocessor, a digital signal processor, a microcontroller, or a similar device.
(141) The input/output device **1120** may be used to input or output data of the system **1100**. The system **1100** may be connected with an external device, e.g., a personal computer (PC) or a network, or exchange data with an external device by using the input/output device **1120**. The input/output device **1120** may be, for example, a touch pad, a keyboard, or a display device.
(142) The memory **1130** may store data for an operation of the controller **1110** or data processed in the controller **1110**. The memory **1130** may include a semiconductor chip diced from the semiconductor substrate according to the disclosure.
(143) The interface **1140** may be a data transmission path between the system **1100** and the external device. The controller **1110**, the input/output device **1120**, the memory **1130**, the interface **1140**

may communicate with one another through a bus **1150**.

(144) While one or more aspects of the disclosure has been particularly shown and described with reference to exemplary embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. A method of dicing a semiconductor wafer, the method comprising: providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing region provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer provided on the active surface across at least a portion of the adjacent integrated circuit regions and the dicing region; forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region; propagating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate; and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer, wherein the method comprises forming a plurality of vertical metal structures directly on the metal shield layer, the plurality of vertical metal structures extending in a minor-axial direction that is perpendicular to the active surface, wherein, prior to the dicing, the metal shield layer is a continuous thin flat plate-type structure having a bottom side surface in direct contact with an upper surface of the semiconductor substrate, and all other side surfaces in direct contact with a semiconductor device layer formed on the semiconductor substrate, except for a portion of one of the side surface contacting the plurality of vertical metal structures, and wherein the plurality of vertical metal structures continuously extend in the minor-axial direction through a plurality of dielectric layers and metal wiring layers.
2. The method of claim 1, wherein, the forming of the modified layer by irradiating the laser comprises preventing a spot generated due to leakage or scattering of the laser from spreading to the adjacent integrated circuit regions using the metal shield layer.
3. The method of claim 1, wherein, in the providing of the semiconductor substrate, a bottom surface of the metal shield layer is arranged to directly contact the active surface.
4. The method of claim 1, wherein the adjacent integrated circuit regions include a first integrated circuit region and a second integrated circuit region that are adjacent to each other, the metal shield layer comprises a first metal shield layer corresponding to the first integrated circuit region and a second metal shield layer corresponding to the second integrated circuit region, the first metal shield layer and the second metal shield layer are spaced apart from each other by a first interval, and the first interval is less than a width of the dicing region.
5. The method of claim 4, wherein each of the first metal shield layer and the second metal shield layer includes a major axis that is parallel to the active surface and a minor axis that is perpendicular to the active surface, and a ratio of a length of each of the first metal shield layer and the second metal shield layer in the major axis to a length of each of the first metal shield layer and the second metal shield layer in the minor axis is about 50:1 to about 200:1.
6. The method of claim 1, wherein, in a planar view, the metal shield layer is continuous across the adjacent integrated circuit regions with the dicing region therebetween, and the crack penetrates the metal shield layer located in the dicing region.
7. The method of claim 6, wherein the metal shield layer covers the active surface corresponding to the dicing region.
8. The method of claim 1, wherein, in the providing of the semiconductor substrate, the metal shield layer includes single metal, and a melting point of material forming the metal shield layer is higher than about 600° C.

9. The method of claim 1, wherein the semiconductor substrate is provided to include a plurality of metal vertical structures for inducing propagation of the crack are arranged on the metal shield layer.
10. The method of claim 9, wherein, in a planar view, the direction of the crack and a major-axial direction of each of the plurality of metal vertical structures are parallel to each other.
11. The method of claim 1, wherein the plurality of dielectric layers and metal wiring layers form a stacked layer including a plurality of inter-layer insulating film and a plurality of metal wires alternatively arranged.
12. A method of dicing a semiconductor wafer, the method comprising: providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing region provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer formed on the active surface across at least a portion of the integrated circuit regions and the dicing region; forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region; propagating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate; and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer, wherein, in a cross-sectional view, the metal shield layer includes a first metal shield layer and a second metal shield layer with a space region therebetween in a location where the crack propagates, each of the first metal shield layer and the second metal shield layer includes a major axis that is parallel to the active surface and a minor axis that is perpendicular to the active surface in the cross-sectional view, and a length of the major axis is about 50 μm to about 100 μm, and a length of the minor axis is about 0.5 μm to about 1 μm, wherein the method comprises forming a plurality of vertical metal structures directly on the metal shield layer, the plurality of vertical metal structures extending in a minor-axial direction that is perpendicular to the active surface, and wherein, prior to the dicing, the metal shield layer is a continuous thin flat plate-type structure having a bottom side surface in direct contact with an upper surface of the semiconductor substrate, and all other side surfaces in direct contact with a semiconductor device layer formed on the semiconductor substrate, except for a portion of one of the side surface contacting the plurality of vertical metal structures, and wherein the plurality of vertical metal structures continuously extend in the minor-axial direction through a plurality of dielectric layers and metal wiring layers.
13. The method of claim 12, wherein, the forming of the modified layer by irradiating the laser comprises preventing a spot generated due to leakage or scattering of the laser from spreading to the adjacent integrated circuit regions using the first metal shield layer and the second metal shield layer.
14. The method of claim 12, wherein, the adjacent integrated circuit regions include a first integrated circuit region and a second integrated circuit region that are adjacent to each other, and in the forming of the semiconductor chips, in a plane view, the first metal shield layer has a rectangular shape arranged along a circumference of the first integrated circuit region, and the second metal shield layer has a rectangular shape arranged along a circumference of the second integrated circuit region that is adjacent to the first integrated circuit region.
15. The method of claim 12, wherein, in the providing of the semiconductor substrate, the first metal shield layer and the second metal shield layer have substantially same shape and same material as each other.
16. The method of claim 15, wherein the first metal shield layer and the second metal shield layer include aluminum.